Section: Senior **TOPIC: Qualitative Analysis** **Date: 09-07-2020**

ONLY ONE ANSWER IS CORRECT

1. Precipitate of $PbSO_4$ is soluble in :
(A) ammonium acetate (B) dilute HCl (C) dilute H_2SO_4 (D) none

2. Nessler's reagent is :
(A) K_2HgI_4 (B) K_2HgI_4 + KOH (C) K_2HgI_2 + KOH (D) K_2HgI_4 + KI

3. Ammonia/ammonium ion gives yellow precipitate with :
(A) H_2PtCl_6 (B) $HgCl_2$ (C) $Na_3[Co(NO_2)_6]$ (D) (A) and (C) both

4. The $K_{s.p.}$, for HgS, Ag_2S and PbS are 10^{-31}, 10^{-45} and 10^{-50} respectively. The solubilities are in the order :
(A) HgS > Ag_2S > PbS (B) HgS < PbS < Ag_2S (C) PbS > Ag_2S > HgS (D) Ag_2S > HgS > PbS

5. Cu^{2+} and Ag^+ are both present in the same solution . To precipitate one of the ions and leaves the other in solution, add
(A) H_2S (aq) (B) HCl (aq) (C) HNO_3(aq) (D) NH_4NO_3(aq)

6. Consider the following observation :
M^{n+} + HCl (dilute) $\longrightarrow$ white precipitate $\xrightarrow{\Delta}$ water soluble $\xrightarrow{CrO_4^{2-}}$ Yellow precipitate.
The metal ion M^{n+} will be :
(A) Hg^{2+} (B) Ag^+ (C) Pb^{2+} (D) Sn^{2+}

7. Yellow ammonium sulphide solution is a suitable reagent for the separation of :
(A) HgS and PbS (B) PbS and Bi_2S_3 (C) Bi_2S_3 and CuS (D) CdS and As_2S_3

8. A metal chloride solution on mixing with K_2CrO_4 solution gives a yellow precipitate soluble in aqueous sodium hydroxide. The metal may be
(A) Mercury (B) Zinc (C) Silver (D) Lead

9. Which of the following is insoluble in dil. HNO_3 but dissolves in aquaregia?
(A) HgS (B) PbS (C) Bi_2S_3 (D) CuS.

10. When small amount of $SnCl_2$ is added to a solution of Hg^{2+} ions, a silky while precipitate is obtained. The silky white precipitate is due to the formation of :
(A) Hg_2Cl_2 (B) $SnCl_4$ (C) Sn (D) Hg

11. When excess of dilute NH_4OH is added to an aqueous solution of copper sulphate an intense blue colour is obtained. This is due to the formation of :
(A) $CuSO_4$ (B) $Cu(OH)_2$ (C) $[Cu(NH_3)_4]^{2+}$ (D) $(NH_4)_2SO_4$

12. When NH_4Cl is added to a solution of NH_4OH :
(A) The dissociation of NH_4OH increases (B) The concentration of OH^- increases
(C) The concentrations of both OH^- an NH_4^+ increase
(D) The concentration of OH^- ion decreases.

13. The solution of sodium meta aluminate on diluting with water and then boiling with ammonium chloride gives:
(A) $[Al(H_2O)_5OH]^{2+}$ (B) $AlCl_3$ (C) $Al(OH)_3$ (D) $NaAl(OH)_4$

14. Intense blue precipitate of $Fe_4[Fe(CN)_6]_3$ and sodium hydroxide solution when mixed gives :
(A) soluble prussian blue (B) reddish–brown precipitate
(C) deep–red colouration (D) turnbull's blue

15. $Fe(OH)_3$ and $Cr(OH)_3$ precipitates are completely separated by :
(A) Aq. NH_3 (B) HCl (C) $NaOH/H_2O_2$ (D) H_2SO_4

16. A metal salt solution when treated with dimethyl glyoxime and NH_4OH gives a rose red complex. The metal is -

(A) Ni (B) Zn (C) Co (D) Mn.

17. An aqueous solution of colourless metal sulphate M, gives a white ppt. with NH_4OH. This was soluble in excess of NH_4OH. On passing H_2S through this solution a white ppt. is formed. The metal M in the salt is

(A) Ca (B) Ba (C) Al (D) Zn

18. Match the colour of the precipitates with their respective compounds (molecular formula) obtained in the analysis of different cations.

Column I	Column II
(A) White crystalline precipitate	(p) $K_3[Co(NO_2)_6]$
(B) Reddish brown precipitate	(q) $Cr(OH)_3$
(C) Yellow precipitate	(r) $Fe(OH)_3$
(D) Green precipitate	(s) $PbCl_2$

Comprehension 1

Aqueous solution of 'A' $\xrightarrow{H_2S\,(g)}$ Black precipitate 'B', soluble in 50% HNO_3 forming 'C'.

Aqueous solution of 'A' $\xrightarrow{NH_3 \text{ solution}}$ White precipitate dissolves in hydrochloric acid but on dilution with water again white turbidity appears 'E'.

'C' $\xrightarrow{\text{Alkaline } Na_2SnO_2}$ Black precipitate 'D'

Moreover, the salt 'A' on heating with solid $K_2Cr_2O_7$ and concentrated H_2SO_4 produces deep red vapours which dissolve in sodium hydroxide solution forming a yellow solution. This yellow solution gives yellow ppt. with $Ba(NO_3)_2$ solution.

On the basis of the aforesaid characteristic information answer the following questions :

19. Acidified solution of 'A', on treatment with KI gives black precipitate 'F' which dissolves in excess of reagent forming the compound 'G'. The chemical composition of 'G' is :

(A) $[HgI_4]^{2-}$ (B) $[PbI_4]^{2-}$ (C) $[BiI_4]^{-}$ (D) None of these

20. The black ppt. 'F' on heating with water produces :

(A) $Hg(OH)_2$ (B*) BiOI (C) BiO.OH (D) CuO.OH

21. Which of the following statement is incorrect.

(A) The black ppt. 'D' is of bismuts. (B) The black ppt. 'D' is of Hg + $Hg(NH_2)NO_3$.

(C) The white turbidity 'E' is of BiOCl. (D) Both (A) and (C).

22. Select the correct statement.

(A) Aqueous solution of 'A' reacts with $AgNO_3$ solution to give white precipitate which turns yellow on treatment with sodium arsenite.

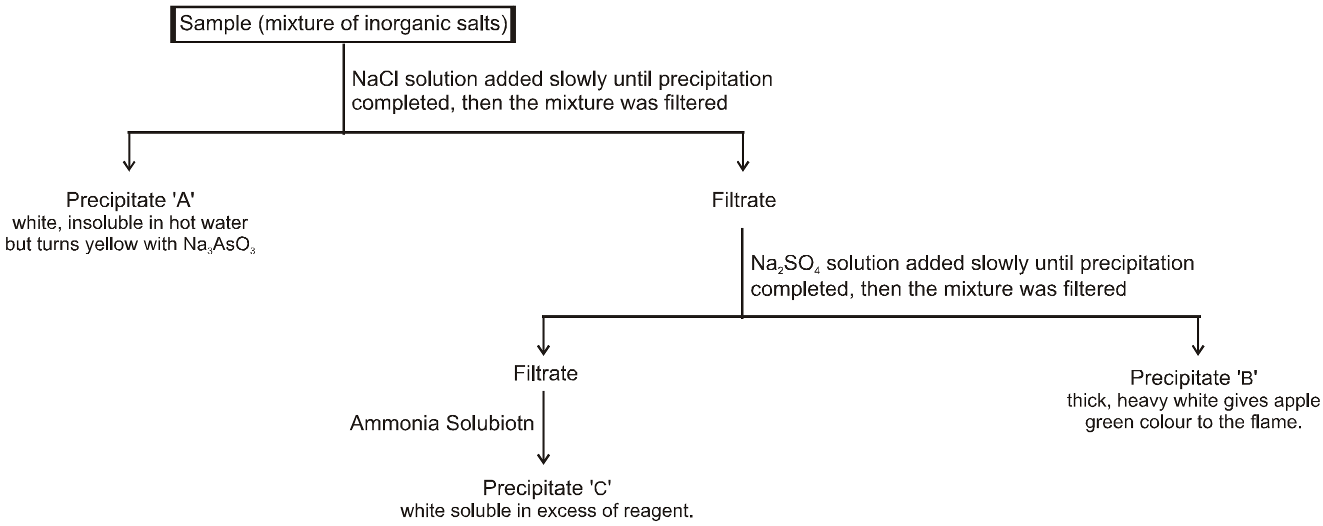
(B) The white turbidity 'E' turns yellow on boiling.

(C) (A) and (B) both.

(D) None of these

Comprehension 2

A student was given a sample of colourless solution containing three cations and was asked to identify the cations. Student carried out a series of reactions as given below :



23. Precipitates 'A', 'B' and 'C' are respectively :

(A) $Al(OH)_3$, $BaSO_4$ and AgCl
(B) AgCl, $BaSO_4$ and $Zn(OH)_2$
(C) AgCl, $ZnSO_4$ and $Ca(OH)_2$
(D) $ZnCl_2$, $BaSO_4$ and $Al(OH)_3$

24. Addition of NH_3 (aq) will dissolve :

(A) precipitate 'A' (B) precipitate 'B' (C) precipitate 'C' (D) None of these

25. Which of the following statement is correct ?

(A) Precipitate 'C' gives Rinmann's green test.
(B) Precipitate 'B' is appreciably soluble in boiling concentrated H_2SO_4.
(C) Precipitate (A) soluble in ammonia solution gives back precipitate with dil. HNO_3.
(D) All of these.

KEY

1.	A	2.	B	3.	D	4.	A	5.	B	6.	C	7.	D
8.	D	9.	A	10.	A	11.	C	12.	D	13.	C	14.	B
15.	C	16.	A	17.	D								
18.	(A–s), (B–r), (C–p), (D–q).												
19.	C	20.	B	21.	B	22.	A	23.	B	24.	A	25.	